Fran Cortés

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Experience

AI Researcher Visitor Cambridge Uni (CaMLSys) *Cambridge, UK* **09/2025 – 09/2025**

- Academic research in agentic red teaming, targeting LLMs, focused on identifying vulnerabilities and evaluating model robustness, collaborating with Nic Lane's group (CaMLSys Lab, University of Cambridge); collaboration ongoing. Contributed to the Flower federated learning framework: github.com/flwrlabs/flower/issues/6039

AI Researcher Engineer Visitor IBM *Dublin, Ireland* **05/2025 – 08/2025**

- Contributed to academic research and the development of a red teaming system targeting LLMs, focusing on identifying vulnerabilities and evaluating model robustness); collaboration ongoing. github.com/IBM/ares/commit/c0e0b69f92a865cdf3d51b792835a511d028a6f5

AI | Cybersecurity Researcher Universidad de Murcia *Murcia, Spain* **10/2024 - Current**

- AI Researcher Engineer focused on Federated Learning and Cybersecurity applied to IoT, leveraging Large Language Models and advanced AI techniques for vulnerability identification and penetration testing.
- Contributed to academic research and the development of secure, privacy-preserving machine learning systems in collaboration with interdisciplinary teams, under the supervision of Prof. Antonio Skármeta and Prof. José Luis Hernández Ramos.

Specialist Software Engineer Vodafone *Málaga, Spain* **09/2023 – 10/2024**

- Developed microservices using Java Spring Boot, MongoDB, Apache Kafka and Docker, while adhering to agile methodologies and industry best practices while deploying new features into production.
- Led the technical hiring process for new graduate engineers, including designing technical scenarios and interview questions.
- Mentored our internship engineers to create a final project, simplify the DB budget management process across portfolios using power app while building a chatbot as an extension.
- Proposed and designed using figma a frontend application to track our service orders and help to clarify our demos.

Deep Learning Engineer Lead Bemyvega *Murcia, Spain* **12/2022 - 05/2023**

- Led the AI team in developing real-time transcription systems using NLP and deep learning techniques, including LLMs. Analyzed and selected appropriate models for each use case, resulting in high-quality transcriptions.

AI Software Engineer | CEO CO-Founder MiSintaxis *Murcia, Spain* **09/2019 - 09/2023**

- Led the creation, design, development and strategy of the number one app in the world of syntax teaching, with more than 85,000 downloads all around the world.
- Applied Natural Language Processing techniques for Spanish language processing. Conducted fine-tuning of the Bloom and GPT2 model on the Ancora corpus, resulting in impressive F1 metrics for Spanish constituency parsing. Using Hugging Face, PyTorch, Python and AWS to deploy it. Design and implement an architecture based on AWS services.
- Designed and implemented scalable APIs and background workers to develop this mobile app.
- Used android studio with Java and XML at first and then led the migration to flutter in order to create a multiplatform app.
- Recruited 1 designer, 3 software engineers.
- Used the lean methodology to create a minimum viable product and implemented the agile methodology with SCRUM using Jira.
- Managed the business side, including legal issues, branding, recruiting, fundraising, external communications and networking.

Projects

- **Movie Recommendation System**: Created with my colleague a hybrid system using content-based, collaborative-based and Knowledge-based filtering using Java (Github Link) (2/2022)
- **TURBINE**: Before going to college, I created a 3D mobile video game for Android using Unity 3D with C# and Blender. The game was available in the play store for free. Link to YouTube Gameplay (07/2018)
- **ANALIZANDO**: Created a syntax teaching app in high school at the age of 15 in order to help students prepare for exams. Used Android Studio and developed it with Java and XML. The app was available in the Play Store, and it had more than 30,000 downloads (09/2016)

Education

Master's Degree in Artificial Intelligence Universidad de Murcia *Murcia, Spain* **09/2025 - 07/2026**

- Major in Artificial Intelligence. Overall GPA of 9.14/10. Master's thesis — an agentic framework for the automated red-teaming of LLMs — graded 9.9/10 (Sobresaliente).

Bachelor of Computer Engineering with Honors Universidad de Murcia *Murcia, Spain* **09/2019 - 07/2023**

- Major in Computer Engineering. GPA of 8.6, this is in the top 1% of the university.

Erasmus-Computer Engineering Politechnika Wrocławska *Wroclaw, Poland* **09/2021 - 02/2022**

- Major in Computer Engineering. Studied abroad in Poland for one semester. All classes were in English.

Research

- **Cortés-Delgado, F. J.** and Rodríguez, M., et al. (2026). ContextualJailbreak: Evolutionary Red-Teaming via Simulated Conversational Priming. arXiv:2605.02647. Black-box evolutionary search over simulated multi-turn primed dialogue, guided by a graded 0-5 harm score from a two-level judge. Across 50 HarmBench behaviors reaches 100% ASR on gpt-oss:20B, qwen3-8B, and llama3.1:70B, and 90% on gpt-oss:120B; the 40 most harmful attacks transfer without adaptation to closed frontier models (90% gpt-4o-mini, 70% gpt-5, 70% gemini-3-flash), revealing a pronounced provider-level asymmetry in alignment robustness.
- **Cortés-Delgado, F. J.**, Mármol Campos, E., Hernández-Ramos, J. L., Skarmeta, A., Latif, S., Djenouri, D., Krenn, S., Puiu, A., & Vizitiu, A. (2026). Enhancing security and privacy in federated learning for distributed systems: The REMINDER approach. In Proceedings of the Global IoT and Edge Computing Summit (GIECS 2025) (pp. 57–70). Springer. doi.org/10.1007/978-3-032-09555-8_4

Skills

- Natural Language Processing | Deep Learning | Hugging Face | PyTorch | Python
- Java Spring Boot | Docker | JavaScript | HTML | CSS | C++ | C | MongoDB | MySQL | AWS | Git | SCRUM | Unit Testing | Cucumber | BDD
- Management | Microservices | Artificial Intelligence | Full-Stack
- English, Spanish, French – All professional proficiency or above

Awards

- **Extraordinary Final Degree Project Award**: Receive recognition for my final project. Applied Natural Language Processing techniques for Spanish language processing. Conducted fine-tuning of different LLMs. Design and implement an architecture based on AWS services.
- **Extraordinary High School Award**: Received a recognition for passing every subject of my final year of high school with Honors.
- **First place in** multiple startup competitions and regional contests with my startup, MiSintaxis.
- **Awarded scholarships** to take courses in business administration and management.

Others

- Volunteered as a mentor for international students in my city.
- Performed Tech Talks in different universities and in different high schools.